Vision Transformers (ViTs)

How attention mechanisms are reshaping the future of computer vision — from medical imaging to autonomous systems and generative AI.

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The Limits of Traditional CNNs

Local Receptive Fields

CNNs process pixels through small, fixed kernels — making it difficult to capture long-range spatial dependencies without stacking many layers.

Scaling Bottlenecks

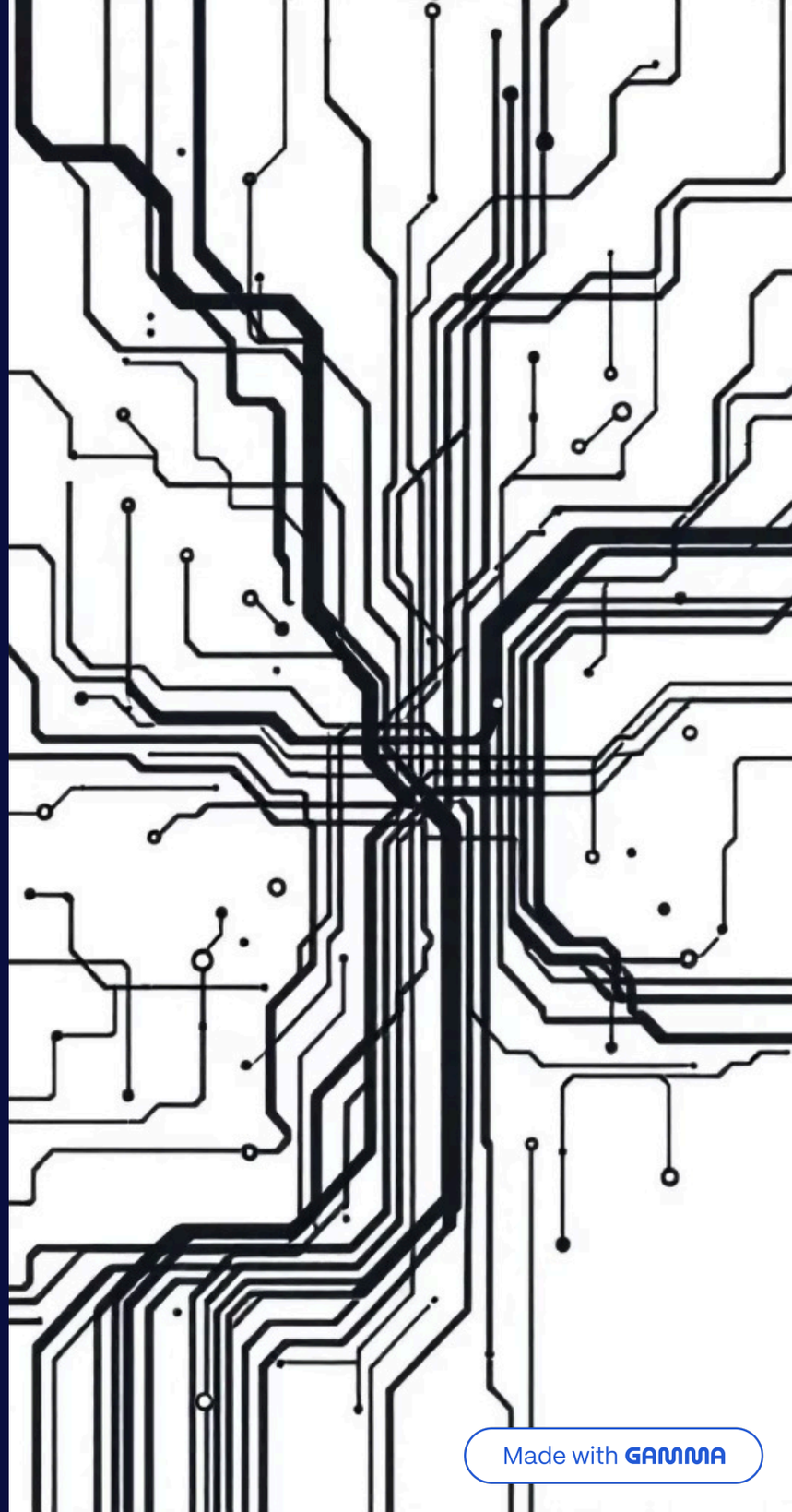
Deeper CNNs demand quadratically growing compute and carefully tuned architectures to scale effectively across diverse tasks.

Inductive Bias

Translation equivariance and locality are baked in — useful, but limiting when global context matters, such as in scene understanding.

The Core Problem

CNNs excel at texture and local patterns, but struggle to model the **global structure** of an image without significant architectural overhead. Transformers address this directly.



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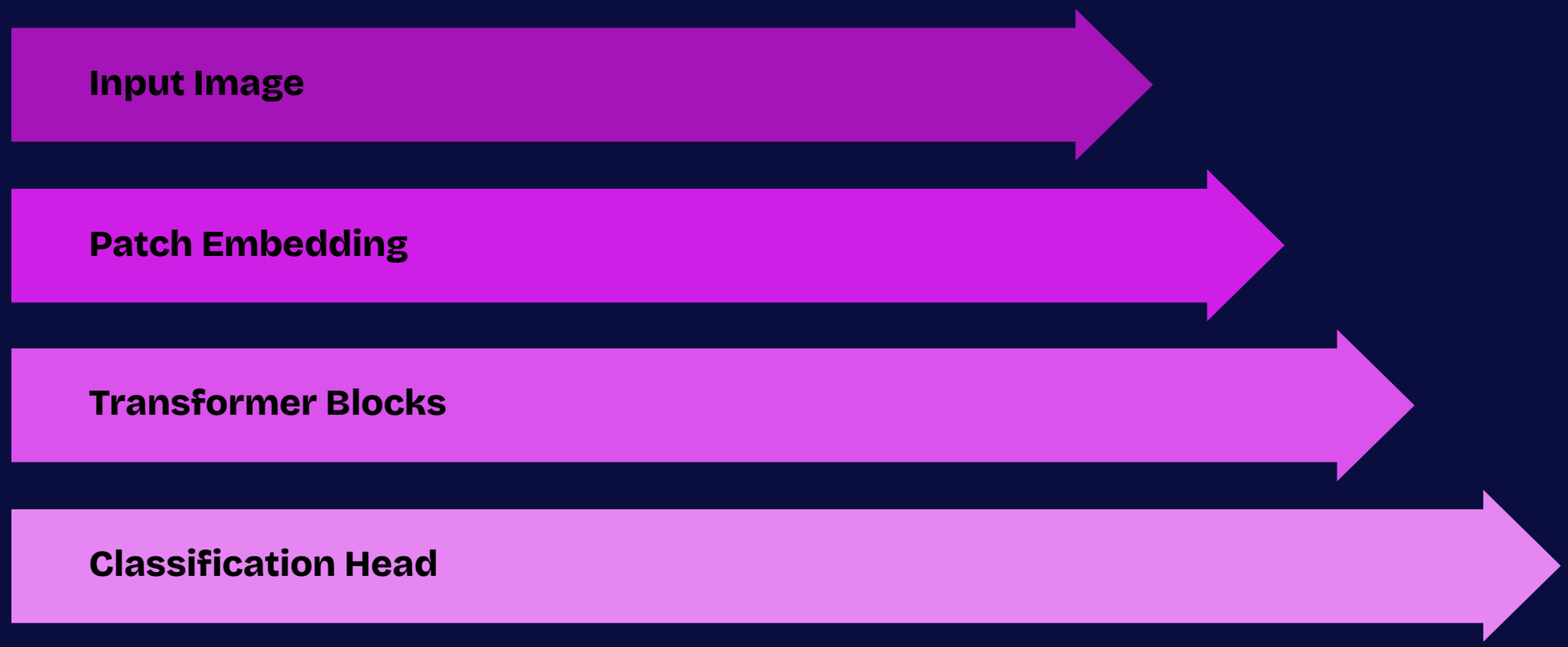
What Is a Vision Transformer?

Introduced by Dosovitskiy et al. (2020), ViT applies the **Transformer architecture** — originally designed for NLP — directly to image data, replacing convolutions with self-attention.



Rather than scanning pixels with a kernel, ViT treats an image as a **sequence of patches** — much like words in a sentence — enabling the model to attend globally from the very first layer.

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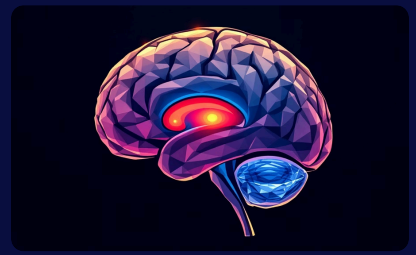
Applications: Healthcare & Medical Imaging



Tumour Detection

ViTs trained on MRI and CT datasets localise anomalies with superior sensitivity, leveraging global context to distinguish subtle pathological features.

Histopathology

Whole-slide image analysis benefits from ViT's ability to model long-range tissue structures — critical for cancer grading and prognosis.

Retinal Analysis

Models like TransUNet and Swin-UNet achieve state-of-the-art segmentation in diabetic retinopathy screening and optic disc mapping.

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Applications: Autonomous Driving

Perception at Scale

Self-driving systems require robust, real-time understanding of dynamic, cluttered scenes. ViTs bring several key advantages over CNN-based pipelines:

Industry Adoption

Tesla's FSD, Waymo, and Mobileye have all shifted towards Transformer-based backbone architectures for their perception stacks.

→ 3D Object Detection

Models like DETR and BEVFormer use cross-attention to fuse multi-camera views into a unified bird's-eye-view scene representation.

→ Lane & Obstacle Segmentation

Global attention captures long-range road context — essential for detecting partially occluded pedestrians and lane markings.

→ Sensor Fusion

Transformer encoders naturally unify heterogeneous inputs from cameras, LiDAR, and radar through a shared attention interface.

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Applications: Generative AI

Stable Diffusion

The U-Net denoising backbone incorporates Transformer attention blocks to model global semantic coherence during image synthesis — enabling photorealistic, prompt-conditioned generation.

Segment Anything Model (SAM)

Meta's SAM uses a ViT-H backbone to produce zero-shot segmentation masks from arbitrary prompts, demonstrating unprecedented generalisation across domains.

Vision-Language Models

CLIP, ALIGN, and Flamingo use ViT encoders to align visual and textual representations — powering image captioning, visual QA, and multimodal reasoning.

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ViT Advantages: A Summary

Global Receptive Field

Self-attention operates across all tokens from layer one — no need to stack layers to capture distant dependencies.

Scalability

ViTs scale predictably with data and compute. Larger models consistently achieve higher performance without architectural redesign.

Transfer Learning

Pre-trained on large datasets (ImageNet-21k, JFT-300M), ViTs transfer effectively to specialised domains with minimal fine-tuning.

Multimodal Flexibility

The token-based interface unifies vision and language — enabling seamless integration into multimodal architectures without architectural patches.

Key limitation: ViTs are data-hungry and computationally expensive at high resolutions. Efficient variants — Swin, DeiT, and PVT — address these trade-offs through hierarchical or sparse attention strategies.

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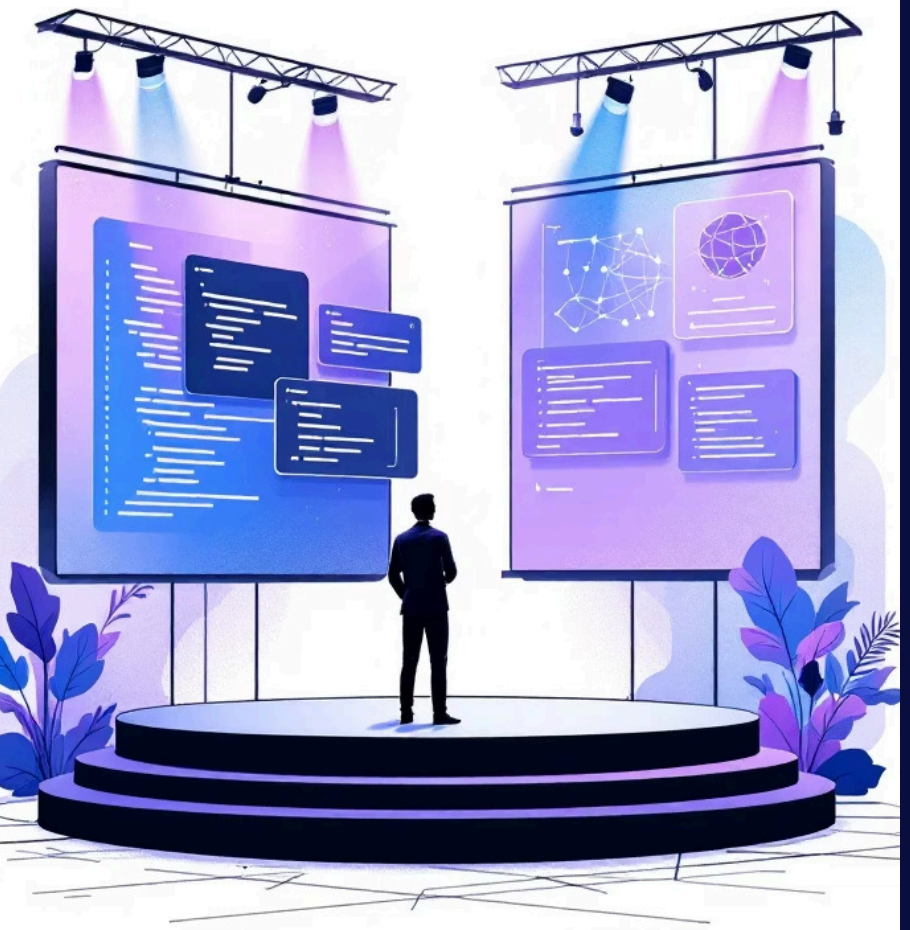

Live Demo Time!

A hands-on walkthrough of a Vision Transformer in action. We will explore patch embeddings, attention maps, and inference on real-world images.

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